

FVT signal study and analysis results-binding code

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First of all, it needs to be clear to you that this program and related analysis content have been clearly explained in the author's master's thesis (has not been published online, has been submitted to the school and library system) (there are some small annotation errors in the paper, but do not affect the overall analysis results).

In order to test the universality of the signal analysis tool, the relevant code and instructions (signal processing, how to see the signal, etc.) are detailed. Please contact the author to quote in the application, respect the author's discovery and innovation points, once found the discovery or analysis of the author as your innovation point, without explanation and quoting, will report the academic misconduct or notify the relevant processing institutions.

Welcome, we will share and communicate with the author about your analysis results and the results of the map, learn together, grow together and make progress together.

Below, the relevant content in the code base, how to analyze and view, to explain in detail.

The technical points, the author thinks (only a personal opinion), is a very useful fault diagnosis tool. Professor Han Jie of Zhengzhou University and his team have done a lot of research, and related papers and books have a lot of explanations on its theory and application.

However, the author's idea has always been: because the time domain is difficult to provide researchers with more direct, clear and rich characteristic information, at the same time, especially for the actual signals, there are many interference factors, it is even more difficult to analyze in the time domain. Therefore, many relevant studies and staff turn to the frequency domain for analysis and research. Therefore, that is

to say, if some information about the frequency domain can be clearly expressed in the time domain, then the frequency domain is only a supplementary tool to assist the time domain analysis, and is no longer strongly dependent on the characteristics provided by the frequency domain. After all, the frequency domain is a way of expression of the time domain. Based on this idea, the full vector spectrum is a homologous signal fusion technology, which can enrich the signal and has excellent performance in the spectrum. Therefore, we can conduct research and analysis based on this technology.

Therefore, relevant researchers can learn about the full vector spectrum, then, one point is that in the analysis, the author constructed the frequency function, but the direct application of the vector spectrum, not constructing the frequency function, the final of the full vector signal FVT signal result is not big, all this is also because of the characteristics of the frequency domain.

So, go straight to the subject, how to apply it? How to analyze it? This is the author's discovery and innovation, related research results, the author's team will meet you in different application forms and research results. You can also look forward to it, oh!

1、Cycle extraction (vibration signal (including periodic shock information))

Take the bearing data of the Western Reserve University as an example, in the code, the corresponding data set is "12k_Drive_End_B007_0_118.mat".

(1) Step1: Modify code (example below)

```
The clc, clear%%%%%%%% clear history
```

```
road='12k_Drive_End_B007_0_118.mat';%%%%%%%% file name
```

```
Fill = load (load);%%%%%%%%% Get the file
```

```
T = 30000;%%%%%%%%% View data length 122571 in file, try to select  
30000
```

```
And n = 1;%%%%%%%%% Number of cycles
```

```
%% X118_DE_time 122571x1 double  
%% X118_FE_time 122571x1 double  
%% X118_RA_time 122571x1 double  
%% X118_RPM 1795
```

```
yy(:,1)=file.X118_DE_time (1: T * n, 1);%%%%%%%%% channel 1
```

```
yy(:,2)=file.X118_FE_time (1: T * n, 1);%%%%%%%%% channel 2
```

```
Fs = 12000;%%%%%%%%% sampling frequency
```

```
T = (0: length (yy (:, 1)) -1) / Fs;%%%%%%%%% time distribution
```

```
rx=yy(:,1);
```

```
ry=yy(:,2);
```

```
[X, Xc, Xs, XP, Y, Yc, Ys, YP, FAi Ai, Ai, Rx, RX, RP, RL, RS,  
GP] = FullVectorEntropy (rx, ry);%%%%%%%%% FVT Acquisition-> Rx
```

```
fs=Fs;
```

```
figure
```

```
subplot(3,1,1)
```

```
Plot (t, rx);%%%%%%%%% channel 1
```

```
subplot(3,1,2)
```

```
Plot (t, ry);%%%%%%%%% channel 2
```

```

subplot(3,1,3)
plot(t,Rx);%%%%%%%%FVT
The ylim ([-1.1,1.1])%%%%%%%%y-axis coordinate limit for a better
view

```

```

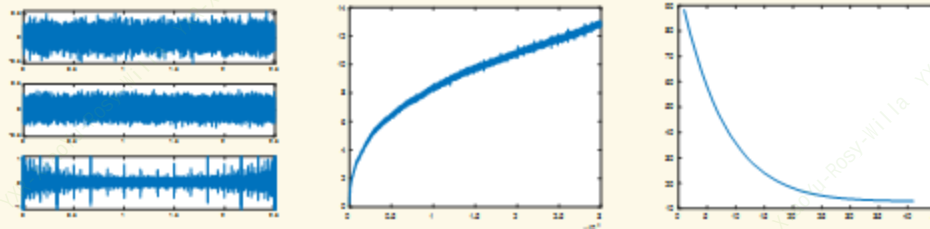
for i = 1:length(yy(:,1))%%%%%%%% Characteristics of FVT
starting position amplitude over time
    rx=yy(1:i,1);
    ry=yy(1:i,2);
    [X,Xc,Xs,XP,Y,Yc,Ys,YP,FaiAi,Ai,Rx,RX,RP,RL,RS,GP]=
FullVectorEntropy(rx,ry);
    startA(1,i)=Rx(1,1);
End

%%%%%%%% Trends of FVT with SNR at the same cycle length
j=1;
for i = -10:30
    rx=awgn(yy(:,1),i,'measured');
    ry=awgn(yy(:,2),i,'measured');
    [X,Xc,Xs,XP,Y,Yc,Ys,YP,FaiAi,Ai,Rx,RX,RP,RL,RS,GP]=
FullVectorEntropy(rx,ry);
    r=Rx(1,1);
    startB(1,j)=r;
    j=j+1;
end

```

```
figure
plot(startA)
figure
plot(startB)
```

(2) Step2: After running, get 3 figure window views

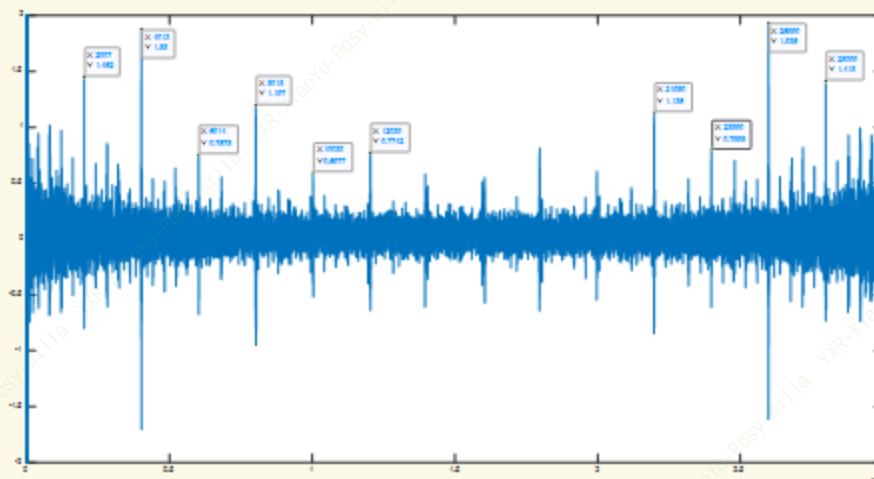


The first figure: the horizontal coordinate is time, the first small figure is the original time domain signal of channel 1; the second small figure is the original time domain signal of channel 2; the third small figure is the FVT signal of channel 1 and channel 2 signal (because the basic theory is the full vector expression is Full Vector, because the author finally get the time signal, therefore, the last corresponding to Time time, so the FVT signal).

Second figure: The abscissa is the data point, and the ordinate is the amplitude of the start position of the FVT signal.

The third figure: the abscissa is the signal to noise ratio, the actual abscissa range is $[-10, 30]$, and the ordinate is Amplitude of the FVT signal start position.

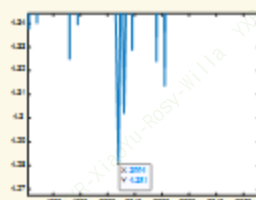
(3) Step3: the extraction period



Enter in the command line window: figure, plot (Rx), ylim ([-2 2])

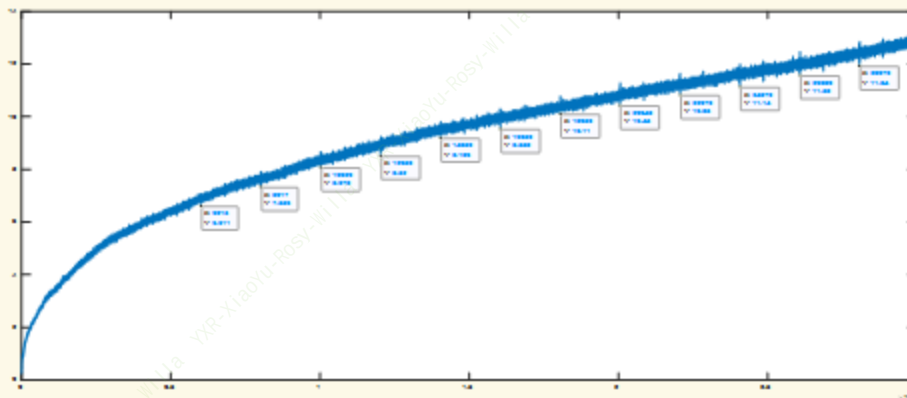
The length of candidate data with periodic information is: 2007 2006
2001 2004 2002 2000

Method 1: (quick method) Open the second figure, and find the amplitude curve position of the starting position of the FVT signal in the range of 2000-2007 data points, and find the minimum value in the range of [2000,2007], which is 2004.



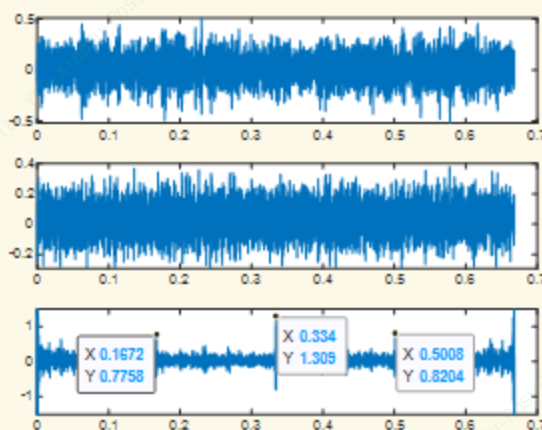
Method 2: (overall method) This method has high requirements on the quality of the signal, but it can also be judged basically. The accuracy is greatly affected by the noise signal quality, as a kind of auxiliary. Taking the minimum cycle drop point, it is found that 2000 2010 2020 2030, at this time, it is necessary to combine FVT signal cycle candidate, and then return to the process of method 1, and finally 2004. Therefore, method 1 is the most direct, the fastest, the second method is the evidence of method 1, which is also the mutual supporting object of FVT cycle

information extraction.



(4) Step4: Extract cycle integrity judgment

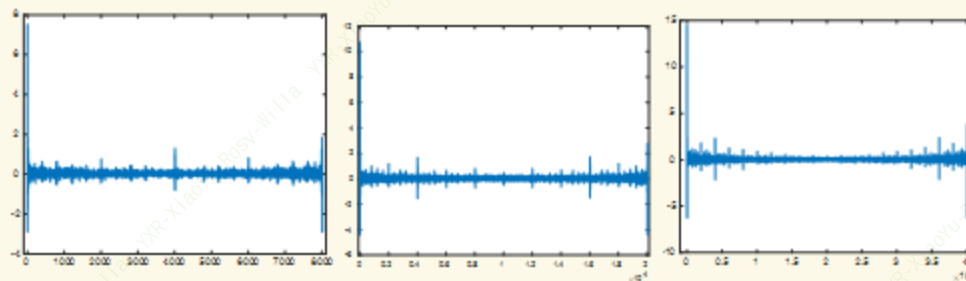
Modify the code to set T to 2004 and 4 to view the Rx-> FVT signal cycle performance results. Found that the periodic pulse in the cycle range is 2004 data points (the difference of 1-2 points is very normal for the actual signal and does not affect the whole), therefore, 2004 is the shortest complete periodic data point length of the signal. The corresponding frequency feature is 5.9881Hz. In fact, after a closer look at the time domain cycle information, we will also find the frequency characteristics of 400 data points length, combined with the characteristics of signal modulation frequency modulation, we can find more modulation information, the author will not say much here.



2、Discrimination of signal quality

Step1: look at the third picture, in the case of other conditions unchanged, the law as the increase of noise ratio, the amplitude of the starting position of FVT signal is smaller and smaller, that, the better the signal quality, the less noise interference, the amplitude of the FVT signal starting position is smaller, this is, the cycle integrity extraction, to pay attention to the amplitude of FVT signal starting position in a falling minimum minimum.

Step2: Look at the first picture, the amplitude difference between the amplitude of the starting position of FVT signal and the impact point of the full cycle position data. When the difference between the two is smaller, the signal quality is better, and the larger the gap between the two, the greater the signal is disturbed. Here, it can be seen that when the number of cycles is increased from 4-> 10-> 20, the FVT signal is increasing, which is consistent with the cycle rise law expressed in the second subgraph. It shows that when the signal length is too long, there is especially interference and the signal quality is poor. Therefore, it is necessary to take the appropriate number of cycles, generally 3 or 4.



3、Evolution signal tracking

Step1: Look at the second picture, the amplitude of the starting position of the FVT signal changes with the increase of the data point. At present, the effect in the prediction is better. Here, because the laboratory has been applied, and the students are already studying, so the author does not give the relevant chart information here. The overall law is that in the process of fault evolution, the vibration signal frequency

modulation is more and more complex, and the amplitude is rising.

4、2-dimensional fusion feature extraction

Step1: Here, you can find out a lot of frequency domain information applications from Professor Han Jie's relevant literature, as the features.

Step2: Here, the more fun is that through a variety of methods of splicing and fusion, you can play a lot of tricks. For example, an output that the author thinks is very interesting (with 20-> 10-> 4 cycles): plot (main vector, secondary vector), we get:

